Final Fermi & Code Problem

Saturday, May 17, 2025 8:38 PM

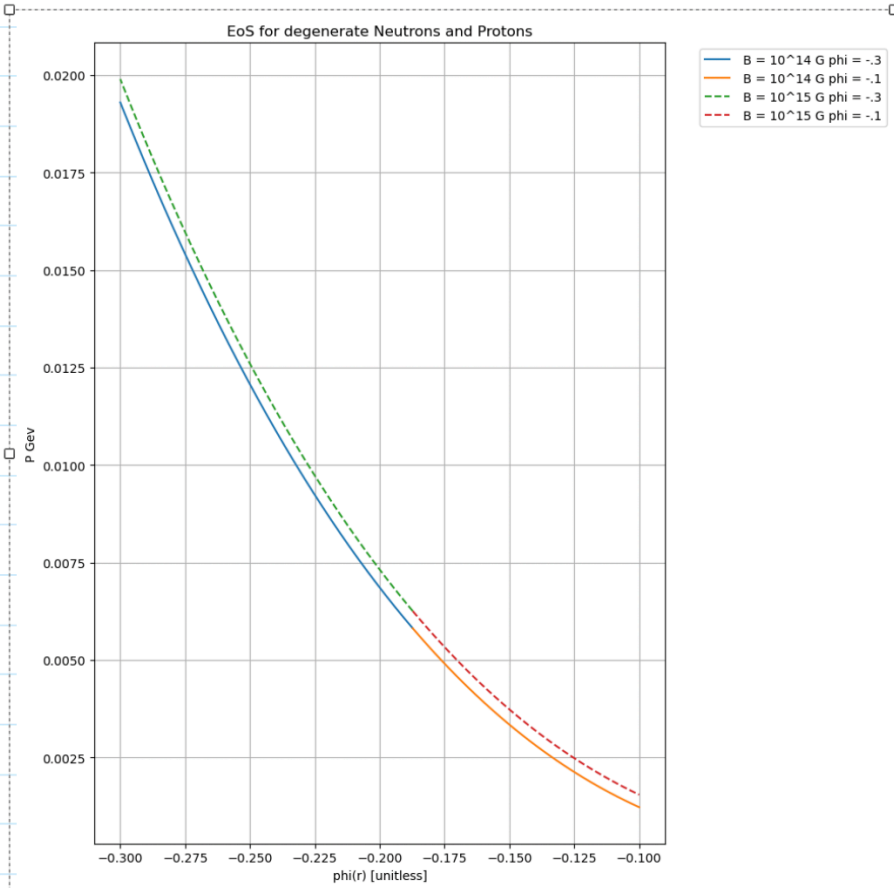
My Fermi/code problem was trying to combine the EoS in my paper into one EoS that is a function or radius. My paper looked at how the magnetic field and employing a curved spacetime effects the EoS for Neutron stars. In the paper they calculate two equations of state: one for degenerate neutrons and another for degenerate charged fermions (a general one that could be used for electrons, protons, or positrons only thing that changes is the mass input). However they never combine the EoS, and we know for neutron stars we expect to see all of these particle. So I wanted to see if I could do that. It was a lot more work than I thought it would be and ultimately a few things were still left as a mystery to me.

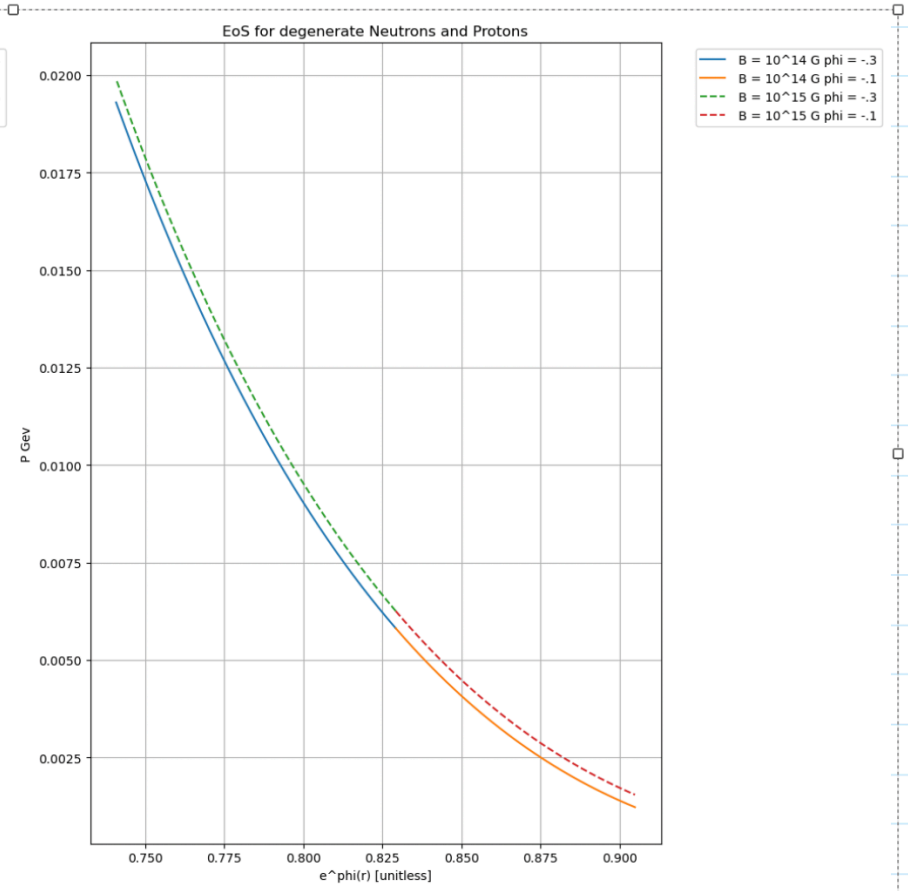
My fermi problem was finding where I should change the EoS depending on the structure of the NS and the math for that is left below.

I formulated my EoS by trying to work backwards to replace the number density dependency with and e^phi dependency. The math for that is below. The main problem with this approach is that the chemical potential is left a mystery because it depends on number density and the number density depends on chemical potential. So I can not create one without creating the other first but they both depend on each other. The way I construct my code: I create an array for phi and then an array for e^phi and then from there construct a number density array based on my formula, but that number density equation depends on the chemical potential. I tried to use physical values that I found in a paper but this left me with a lot of imaginary numbers. So ultimately I decided to set the chemical potential for both neutrons and protons (which is the charged fermion I decided to graph) to 1. This isn't a crazy approximation given the units.

That brings me to my next mystery, I could not for the life of me figure out how to convert units to natural units. I found a few sources that tried to help but in the end I don't think I converted my units right.

My graph looks correct. I expected my EoS's to concatenate together seamlessly because if you compare the graphs in the paper the P vs n for neutrons and charged fermions are pretty similar, so I didn't expect any jumps in my graph. And my pressure dependency looks correct. The way to read the graph is x-axis moves from core to crust: higher negative values means your moving towards the core and the crust is expected to be at phi(R) = -.18. And pressure increase (similarly as in the paper) as you get close to the core.


EoS for degenerate Neutrons and Protons
B = 10^14 G phi = -.3
B = 10^14 G phi = -.1
B = 10^15 G phi = -.3
B = 10^15 G phi = -.1
P Gev
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
−0.300
−0.275
−0.250
−0.225
−0.200
−0.175
−0.150
−0.125
−0.100
phi(r) [unitless]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0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.750
0.775
0.800
0.825
0.850
0.875
0.900
e^phi(r) [unitless]

$$\omega = -c\left(1-\frac{2GM}{c^2R}\right)$$

Fermi Problem:

I have the Φ function for the exterior $e^{2\Phi(R)} = 1-\frac{2GM}{R}$] has the units $\frac{m^2}{s^2}\left[\frac{\text{Length}}{\text{time}}\right]^2$

$$2\Phi(R) = \ln\left[1-\frac{2GM}{R}\right]$$

We're using $M = 1M_\odot$ $R = 10\,Km = 10^4 m$

$G = 7\times10^{-11}\frac{N\,m^2}{Kg^2}$ $M = 1M_\odot = 2\times10^{30}$

so $\Phi(R)$ on the exterior $= -0.17$

$O(GM) = 10^{19}$

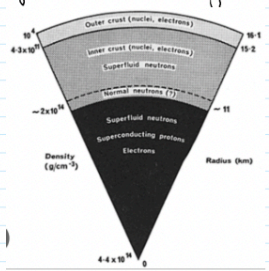
I will graph 2 func's one where the core $\Phi = -1$ another where $\Phi = -3.0$ and do the same mag field stuff they did. Using this NASA diagram for radius guess

$$e^{2\Phi} = \cancel{c^2}^{=1}\left(1-\frac{2GM}{c^2 r}\right)$$

$2r_g$ for $1M_\odot = 1.5\,Km$

r 10 Km

$2\Phi = \ln(.66)$



However Φ is not a direct function of R, I can't just say 1 km difference in radius = 1 unit change in Φ. So I'll guesstimate that difference based on

$$\frac{d\Phi}{dr} = \frac{G(M+4\pi r^3 P_r)}{r(r-2GM)} = \frac{GM}{r^2-2GMr} + \frac{4\pi r^3 P_r}{r^2-2GMr}$$

$$= \frac{1}{r^2} + \frac{r^3}{r^2} = \frac{1}{r^2} + r$$

$$\boxed{\frac{d\Phi}{dr} \propto \frac{1}{r^2} \qquad \Phi \propto \frac{-2}{r^3}}$$

$$8\pi G r^2 P_r = e^{-2\nu}(2r\phi'+1) - 1$$

$$e^{2\nu} = \frac{1}{1-\frac{2GM}{r}}$$

$$P_r = \left[\frac{r-2GM}{r}\cdot(2r\phi'+1) - 1\right]\cdot\frac{1}{8\pi G r^2}$$

$$= \left[\frac{2r^2\phi' - 2GMr\phi'}{r^3} + \frac{r-2GM}{r^3}\right]\cdot\frac{1}{8\pi G}$$

$$\frac{1}{r} + \frac{1}{r^2}$$

For 16 Km NS

For 10 Km NS

$\frac{\Delta r}{R}$

$\frac{1}{16} = .0625$ $\quad \frac{.625}{10} = .0625$ ~ 1

$\frac{5}{16} = .3125$ $\quad \frac{3.125}{10} = .3125$ $\sim 3/2$

$\frac{11}{16} = .6875$ $\quad \frac{6.875}{10} = .6875$ $\sim 6/7$

making EoS a func of radius

$$P_\pm = e^{\phi}\frac{m^4}{48\pi^2}\left[\sqrt{(bn_\pm)^{2/3}+1}\left\{2(bn_\pm)-3(bn_\pm)^{1/3}\right\}+3\,\mathrm{asinh}\left\{(bn_\pm)^{1/3}\right\}\right]$$

$$\rho_\pm = -P_\pm + e^{\phi}\frac{m^4}{6\pi^2}(bn_\pm)\sqrt{(bn_\pm)^{2/3}+1}$$

$$n = n_+ + n_-$$

$$n_\pm = \frac{e^{-3\Phi}}{6\pi^2}\underbrace{\mu_{\pm m}^3}_{=\mu_{sm}} \qquad \mu_{sm} = \sqrt{\mu_s^2 - \bar{m}^2}$$

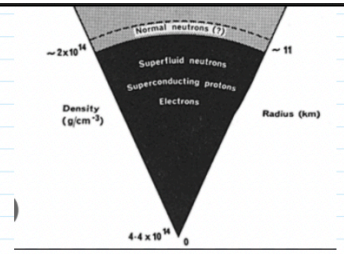
$$\mu_s = \mu + s e^{\Phi}\mu_D B$$

$$b = \frac{6\pi^2}{m^3} \qquad \bar{m} = m e^{\Phi}$$

$$\mu_{sm} = \sqrt{(\mu + s e^{\Phi}\mu_D B)^2 - (m e^{\Phi})^2} \qquad n_\pm = \frac{e^{-3\Phi}}{6\pi^2}\left[(\mu \pm e^{\Phi}\mu_D B)^2 - (m e^{\Phi})^2\right]^{3/2}$$

where $s = \pm$

$$n = \frac{e^{-3\Phi}}{6\pi^2}\left\{\left[(\mu + e^{\Phi}\mu_D B)^2 - (m e^{\Phi})^2\right]^{3/2} + \left[(\mu - e^{\Phi}\mu_D B)^2 - (m e^{\Phi})^2\right]^{3/2}\right\}$$



radius = r and change in s

guesstimate that difference

$$\frac{d\Phi}{dr} = \frac{G(M + 4\pi r^3 P_r)}{r(r - 2GM)} = \frac{\quad}{r^2}$$

$$= \frac{1}{r^2}$$

$$8\pi G r^2 P_r = e^{-2\nu}(2r\phi' + 1)$$

$$e^{2\nu} = \frac{1}{1 - \frac{2GM}{r}}$$

P_r

$$= \left[\right.$$

$$\Phi \propto \frac{-2}{r^3}$$

Outer Crust / charged Fermions: 10 km → 9 km

$\Delta r = 1\,\mathrm{km} / 10^3\,\mathrm{m}$

$$\Delta\Phi \propto -2\left(\frac{1}{10^3} - \frac{1}{9^3}\right)$$

$$= 7\times10^{-4}\,\mathrm{km} / .7\,\mathrm{m}$$

Inner crust: mix of charged Fermions and neutrons

$\Delta r = 3\,\mathrm{km} / 3\times10^3\,\mathrm{m}$

$$\Delta\Phi \propto -2\left(\frac{1}{9^3} - \frac{1}{6^3}\right) = .0065\,\mathrm{km} / 6.5\,\mathrm{m}$$

core: neutrons only $\Delta r = 6\,\mathrm{km} / 6\times10^3\,\mathrm{m}$ thick

$\frac{\Delta r}{R}$	For 16 km NS	For 10 km NS
	$\frac{1}{16} = .0625$	$\frac{.625}{10} = .0625$ ↳ ~1
	$\frac{5}{16} = .3125$	$\frac{3.125}{10} = .3125$ ↳ ~3/2
	$\frac{11}{16} = .6875$	$\frac{6.875}{10} = .6875$ ~ 6/7

How to actually implement this

Φ is an array from $[-.17 \ldots -1.0/-3.6]$ In 100 steps

$r = 10\,\mathrm{km}$ $r = 0\,\mathrm{km}$

At what Φ am I switch eqns?

1st $\Phi = -.1707$ $\qquad$ 2nd $\Phi = -.1772$

Constants

$\mu_{Dn} = -9.66 \times 10^{-27}$ J/T $\qquad -9.66 \times 10^{-27} \frac{J}{T} \cdot \frac{1 eV}{1.6 \times 10^{-19} J} \cdot \frac{1}{10{,}000\ G}$

$m_{n/p} = 1.7 \times 10^{-27}$ Kg $\qquad \mu_{Dn} = -6.0375 \times 10^{-12} \frac{eV}{G} / -9.66 \times 10^{-34} \frac{kg \cdot Km}{s^2\ G}$

$\mu_{Dp} = 1.4 \times 10^{-26} \frac{J}{T} \cdot \frac{1 eV \cdot T}{1.6 \times 10^{-15} J \cdot G} = 8.75 \times 10^{-12} \frac{EV}{G}$

M_neutron = 939.56542194(48) MeV/c2.

Chemical potential for a neutron

↳ The amount of energy that is absorbed | released due to the change of the particle #

from a paper μ for neutrons in pure nuclear matter

$100 - 200$ MeV $\sim 10^8 eV \cdot \frac{1.6 \times 10^{-19} J}{1 eV} \cdot \frac{1 Km}{1000 m} =$

Charged Fermions:

$$n = \frac{e^{-3\Phi}}{3\pi^2} \mu_{m_1}^3 + \frac{eBe^{-\Phi}}{2\pi^2} \mu_m \qquad \mu_m = me^{\Phi}\left[(bn)^{1/3} + \frac{(B/B_c)}{2(bn)^{1/3}}\right]$$

$$\overline{m_\ell}^2 = e^{2\Phi}\left[m^2 + 2(eB)\ell\right] \qquad \mu_{m\ell} = \sqrt{\mu^2 - \overline{m_\ell}^2} \qquad \mu_m = \sqrt{\mu^2 - \overline{m}^2}$$

$$* n = \frac{e^{-3\Phi}}{3\pi^2}\left(\mu^2 - e^{2\Phi}\left[m^2 + 2eB \cdot 1\right]\right)^{\frac{3}{2}} + \frac{eBe^{-\Phi}}{2\pi^2}\sqrt{\mu^2 - (me^{\Phi})^2}$$

$$P = P_0 + P_B$$

$$P_0 = \frac{m^4 e^{\Phi}}{24\pi^2}\left[\sqrt{(bn)^{2/3} + 1}\left\{2bn - 3(bn)^{1/3}\right\} + 3\,\text{asinh}\left\{b(n)^{1/3}\right\}\right]$$ I dependent Part

$$P_B = \frac{m^4 e^{\Phi} B}{12\pi^2 B_c}\left[3\,\text{asinh}\left\{(bn)^{1/3}\right\} - \frac{(bn) + 3(bn)^{1/3}}{\sqrt{(bn)^{2/3} + 1}}\right]$$ magnetic field dependent Part

$b = \frac{3\pi^2}{m^3} \qquad B_c = \frac{m^2}{e}$ $\qquad$ $m \equiv$ mass of proton (im going to use)

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$m \equiv$ mass of proton (im going to use)

$e = 1\,eV$

natural units $c = \hbar = 1$

Quantity	Dimensions		Conversions
	SI Units	Natural Units	
mass	kg	E	1 GeV = 1.8×10^{-27} kg
length	m	1/E	1 GeV^{-1} = 0.197×10^{-15} m
time	s	1/E	1 GeV^{-1}= 6.58×10^{-25} s
energy	$\text{kg}\cdot\text{m}^2/\text{s}^2$	E	1 GeV = 1.6×10^{-10} Joules
momentum	kg·m/s	E	1 GeV = 5.39×10^{-19} kg·m/s
velocity	m/s	-	1 = 2.998×10^{8} m/s (=c)
angular momentum	$\text{kg}\cdot\text{m}^2/\text{s}$	-	1 = 1.06×10^{-34} J·s (=ħ)
cross-section	m^2	$1/\text{E}^2$	1 GeV^{-2} = 0.389 mb = 0.389×10^{-31} m^2
force	$\text{kg}\cdot\text{m}/\text{s}^2$	E^2	1 GeV^2 = 8.19×10^{5} Newton
charge	C=A·s	-	1 = 5.28×10^{-19} Coulomb; e=0.303 or =1.6×10^{-19} C

- mass of a proton/neutron ~ 1 GeV
- mass of an electron ~ 0.5 MeV

- proton/neutron radius ~ 1 fm = 10^{-15} m = 1 **fermi** ~ 1/(200 MeV)
- time for light to cross a proton ~ 10^{-23} s

- highest energy collider (Tevatron) ~ 2 TeV center-of-mass energy
- distance scale probed at Tevatron ~ 1/(2 TeV) ~ 10^{-19} m,
  (this comes from the uncertainty principl